Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

Made with GAMMA

Project Overview

Dataset

3,900 purchases across 18 features

Goal

Uncover spending patterns and customer segments

Outcome

Strategic business recommendations

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Key Data Features

Customer Demographics

- Age, Gender, Location
- Subscription Status

Purchase Details

- Item, Category, Amount
- Season, Size, Color

Shopping Behavior

- Discount Applied
- Promo Code Used
- Previous Purchases
- Frequency, Rating
- Shipping Type

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Data Preparation in Python

01

Data Loading

Imported dataset using pandas

02

Initial Exploration

Checked structure with df.info() and summary statistics with .describe()

03

Missing Data Handling

Imputed missing Review Ratings using median by category

04

Column Standardization

Rename to snake case for readability

05

Feature Engineering

Created age_group and purchase_frequency_days columns

06

Database Integration

Loaded cleaned data into PostgreSQL

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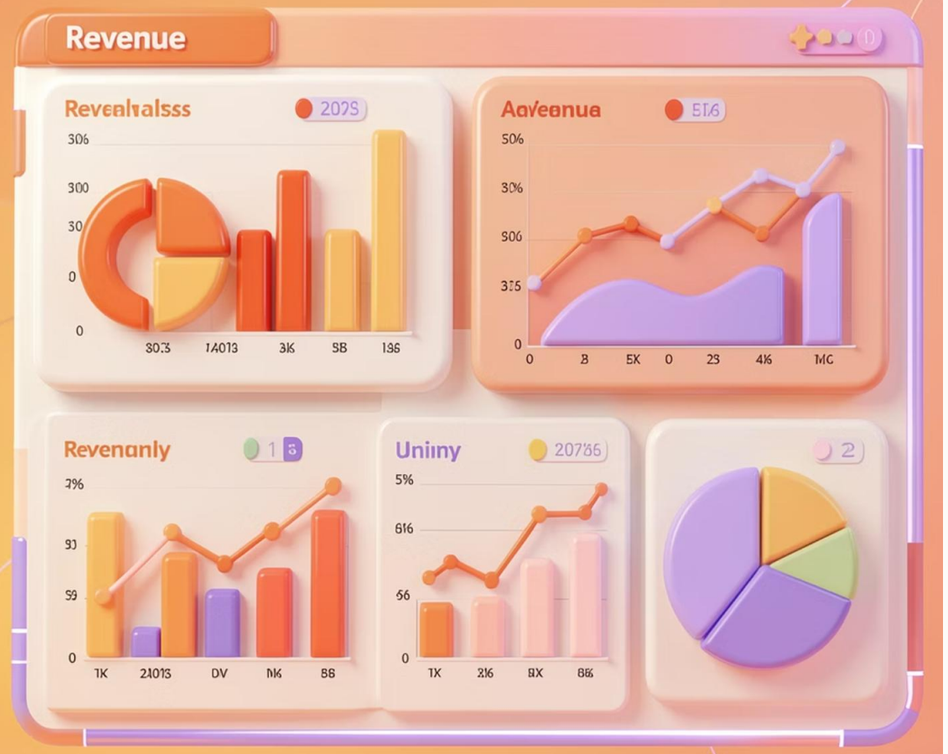

SQL Analysis: Revenue Insights

Revenue by Gender

Compared total revenue generated by male vs. female customers

High-Spending Discount Users

Identified customers who used discounts but spent above average

Top 5 Products by Rating

Found products with highest average review ratings

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SQL Analysis: Customer Behavior

Shipping Type Comparison

Compared average purchase amounts between Standard and Express shipping

Subscribers vs. Non-Subscribers

Compared average spend and total revenue across subscription status

Discount-Dependent Products

Identified 5 products with highest percentage of discounted purchases

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SQL Analysis: Segmentation

Customer Segmentation

Classified as New, Returning, or Loyal based on purchase history

Top 3 Products per Category

Listed most purchased products within each category

Repeat Buyers & Subscriptions

Checked if >5 purchases correlates with subscription likelihood

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Power BI Dashboard

Interactive visualizations presenting key insights from SQL analysis

Customer Behavior Dashboard

Subscription Status

No | Yes

Gender

No | Yes

Category

- Accessories
- Clothing
- Footwear

shipping_type

- 2-Day Shipping
- Express
- Free Shipping
- Next Day Air
- Standard
- Store Pickup

3.9K

Number of Coustomers

$59.76

Average Purchase Amount

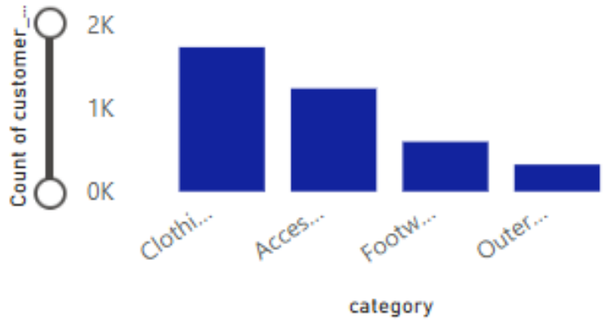
3.75

AVERAGE Review Rating

% of Customer by Subscription Status

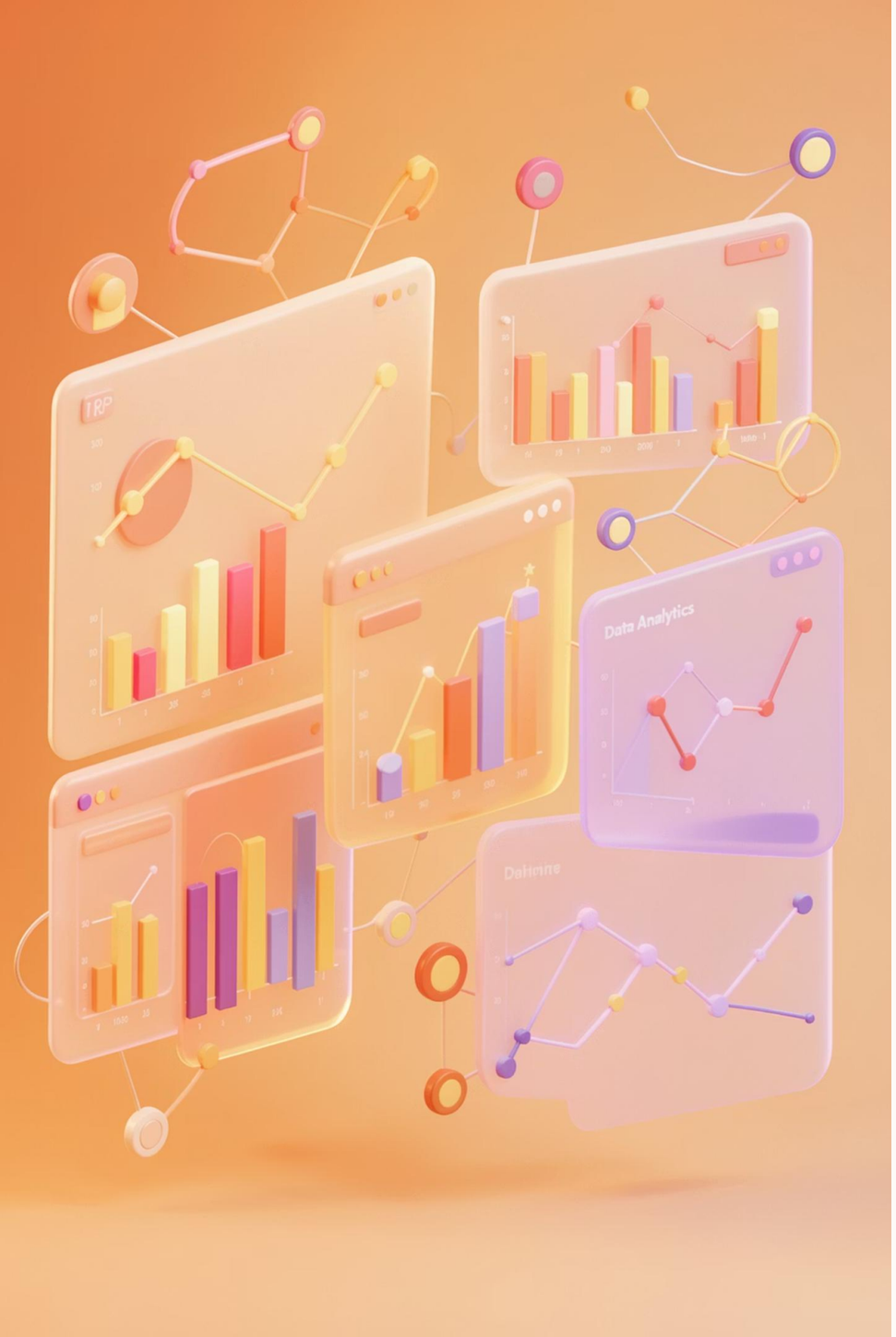
Revenue By Category

Sakes By Category



Revenue of Age Group by age_group

Sales by age_group



Key Findings

3,900

Purchase Records

Analyzed across 18 features

18

Data Features

From demographics to behavior

11

SQL Queries

Answering business questions

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